Notes: Bae, Kang, and Kim (JF, 2002)

Tunneling or Value Added? Evidence from Mergers by Korean Business Groups

Contents

1 One-sentence summary

Bae, Kang, and Kim use Korean chaebol mergers to show that acquisitions by group-affiliated firms often reduce bidder value while increasing the value of other group firms or controlling shareholders, supporting the tunneling view over the value-added view of business groups.

2 Big picture

2.1 Research question

The paper asks whether Korean business groups add value to member firms through internal capital markets and resource sharing, or whether they allow controlling shareholders to transfer wealth from minority shareholders through transactions such as mergers.

2.2 Core empirical setting

The study examines merger announcements by Korean Stock Exchange listed bidders from 1981 to 1997. Korea is useful because chaebols have concentrated control, weak minority shareholder protection, cross-debt guarantees, and extensive internal shareholdings.

2.3 Main result

When a chaebol-affiliated bidder announces an acquisition, its stock price tends to fall. This loss is concentrated among Top 30 chaebol bidders. At the same time, the market value of other firms in the same group often rises, and the controlling shareholder can benefit through holdings in those firms.

2.4 Economic interpretation

The evidence is more consistent with tunneling than with a simple value-added internal-capital-market view. The merger can hurt the listed bidder but help the controlling shareholder by transferring resources toward other group firms or rescuing weak affiliates.

3 Incremental contribution of the paper

3.1 A direct test of business-group benefits versus expropriation

Earlier work highlighted both beneficial internal markets and agency costs in business groups. This paper designs a direct stock-market test that compares the value-added and tunneling hypotheses using merger events.

3.2 Event-study evidence from an emerging market

The paper brings standard corporate finance event-study methods to Korean chaebol mergers, providing a high-frequency market reaction measure of whether acquisitions create or destroy value for bidder minority shareholders.

3.3 Separating bidder shareholders from controlling shareholders

A key contribution is to distinguish the bidder's stock-market loss from the controlling shareholder's overall wealth change across the business group. This distinction is central to tunneling: minority shareholders in one firm can lose while the controller benefits elsewhere.

3.4 Using group affiliation and related-party merger structure

The paper uses chaebol rank, target quality, same-group transactions, cash-flow measures, and ownership structure to identify settings where tunneling is most likely.

3.5 Broader corporate-governance contribution

The paper provides early systematic evidence that controlling shareholders in weak investor-protection environments can use mergers as a mechanism for value transfer.

4 Data and measurement

4.1 Merger sample

The main sample consists of 107 nonfinancial bidders listed on the Korean Stock Exchange between 1981 and 1997. The initial merger announcement date is collected from the *Securities Daily*. Cases where the bidder already owns all shares of the target before the merger are excluded.

4.2 Business group classification

Bidders are classified into Top 30 chaebol bidders, Top 31–50 chaebol bidders, and other bidders. The classification follows the Korea Fair Trade Commission ranking of business groups by aggregate assets.

4.3 Event-study variables

Abnormal returns are computed using the market model with parameters estimated from 200 trading days ending 20 days before the merger announcement. The Korea Composite Stock Price Index is used as the benchmark. The paper reports daily abnormal returns and cumulative abnormal returns over windows such as $(-5, 5)$.

4.4 Key explanatory variables

Important variables include bidder cash flow to assets, prior industry-adjusted stock returns, target financial condition, whether the target is listed, industry relatedness, same-group merger, bidder size, bidder leverage, and ownership concentration by the largest shareholder and other corporations.

4.5 Inside ownership and group-portfolio measures

To study controller incentives, the paper calculates changes in the market value of insider holdings and changes in the value of other firms in the same chaebol group around the merger announcement.

5 Empirical specification and theoretical hypotheses

5.1 Value-added hypothesis

The value-added view predicts that business groups alleviate missing market institutions. Under this view, chaebol mergers should improve bidder value, especially when the group can reallocate capital or managerial resources efficiently.

5.2 Tunneling hypothesis

The tunneling view predicts that mergers can transfer resources from one firm to another for the benefit of the controlling shareholder. The bidder can lose value if it overpays for a weak affiliate or rescues another group firm.

5.3 Event-study regression

The core cross-sectional regression is:

$$CAR_i(-5,5) = \alpha + \beta_1 Top30_i + \beta_2 Top31\_50_i + \Gamma' X_i + \varepsilon_i.$$

A negative β_1 means Top 30 chaebol bidders lose more value around acquisition announcements than other bidders, holding deal and firm characteristics fixed.

5.4 Rescue merger specification

The rescue-merger tests examine whether good chaebol bidders acquire bad targets in the same group:

$$CAR_i = \alpha + \delta Rescue_i + \Gamma' X_i + \varepsilon_i.$$

A negative coefficient for the bidder and a positive effect for other group firms support tunneling through internal rescue.

5.5 Controller-wealth tests

The paper also examines changes in insider wealth and group-portfolio value:

$$\Delta Wealth_{controller} = \Delta Value_{bidderholdings} + \Delta Value_{othergroupholdings}.$$

This shows whether controlling shareholders gain even when bidder minority shareholders lose.

6 Identification and empirical design

6.1 What is identified

The paper does not identify a randomized treatment effect of chaebol membership. Instead, it identifies market reactions to merger announcements and asks whether those reactions match the predictions of value added or tunneling.

6.2 Why event windows matter

Short-window abnormal returns isolate the market's reassessment of bidder value around the merger announcement, limiting contamination from slow-moving macroeconomic conditions or long-run performance trends.

6.3 Cross-sectional prediction tests

The design uses cross-sectional variation in chaebol affiliation, cash flow, ownership concentration, target weakness, and same-group merger status. The tunneling view predicts more negative bidder returns when strong chaebol bidders rescue weak affiliates and when controlling ownership insulates managers.

6.4 Minority versus controller wealth

A central identification idea is that the same event can have opposite implications for different claimants. Bidder shareholders can lose while the controlling shareholder gains through other holdings. This claimant-level distinction is what makes the tunneling hypothesis testable.

6.5 Main limitations

The sample is not randomized, and chaebol bidders differ from nonchaebol bidders in size, leverage, and governance. The paper addresses this with controls and cross-sectional tests, but the results should be read as event-study evidence consistent with tunneling rather than a randomized causal effect.

7 Tables and figures

7.1 Table I: Characteristics of the Top 30 Chaebols, the Top 31–50 Chaebols, and Other Small Chaebols and Independent Firms; Table II: Prediction and Summary Results of Bidder Announcement Returns for Value-Added and Tunneling Hypotheses

Read: Table I documents that the Top 30 chaebols are more diversified, economically larger, more levered, and more connected through cross-debt guarantees and reciprocal ownership. Table II lays out the prediction matrix: value added predicts positive bidder returns, while tunneling predicts negative bidder returns and gains to other group firms or insider holdings.

Economic message: The first table establishes why tunneling is plausible in Korea; the second table defines the empirical predictions that discipline the event-study tests.

Table I

Characteristics of the Top 30 Chaebols, the Top 31–50 Chaebols, and Other Small Chaebols and Independent Firms

This table presents the characteristics of the Top 30 chaebols, the Top 31–50 chaebols, and other small chaebol and independent firms as of tl end of 1997. The stock exchange-related data are obtained from the KSE. The data for chaebols are obtained from the Korea Fair Tra Commission.

Features	Top 30 Chaebols	Top 31–50 Chaebols	Other Small Chaebols and Independent Firms
1. Definition	Thirty largest business groups as ranked by the KFTC in the order of the aggregate assets of all affiliated firms within each group.	Business groups ranked between the Top 31 and 50.	Business groups ranked below the Top 50 and independent firms that do not belong to a business group (i.e., all firms that do not belong to the Top 5 chaebols).
2. Degree of diversification	The Top 30 chaebols operate in many different industries and are broadly diversified. The average number of affiliated firms is 26. The top five chaebols have as many as 62 affiliated firms under their control.	The average number of affiliated firms is 11.	Less diversified and relatively focused.
3. Economic power	The economic power exercised by the Top 30 chaebols is substantial. The percentages of total assets and gross sales that they contribute to listed firms are 62.54% and 72.63%, respectively.	The percentages of total assets and gross sales that they contribute to listed firms are 5.63% and 3.98%, respectively.	The percentages of total assets and gross sales that they contribute to listed firms are 31.82% and 23.38%, respectivel

(a) Table I: characteristics of chaebols and independent firms.

Table II

Prediction and Summary Results of the Bidder Announcement Returns for Value-Added and Tunneling Hypotheses

This table presents prediction and summary results of bidder announcement returns for value-added and tunneling hypotheses. The value-added hypothesis states that business groups in emerging markets add value to their member firms. The tunneling hypothesis states that business groups in emerging markets provide controlling shareholders with an opportunity for wealth transfer from the firm for their own benefit. The sample consists of 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We initially identify the sample from the *Securities Daily*, published by the KSE. We eliminate cases in which the acquiring firm owns all the shares of the acquired firm prior to the merger. "Chaebol bidder" is a firm that belongs to one of the 30 or 50 largest chaebols. "Good chaebol bidder" defines a chaebol firm with the ratio of cash flow to total assets above the sample median. "Bad target" defines a firm with net income or book equity below zero. "Business group" defines a group with at least two member firms. Insider holdings are shares held directly or indirectly by the owner-manager and his/her family. We obtain the market value changes of insider holdings by multiplying the insider holdings by the price change around the announcement date (from −5 to +5).

Panel A: Prediction and Summary Results of Bidder Announcement Returns

Bidder Category	Prediction for Bidder (Other Members of the Chaebol) Announcement Returns		Empirical Results for Bidder (Other Members of the Chaebol) Announcement Returns
	Value-Added	Tunneling	
Chaebol bidder	+	−	−
Good performance (high cash flow or high industry-adjusted excess return)/ chaebol bidder	+	−	−
Concentrated equity ownership (high equity ownership of the largest shareholder and other corporations)/chaebol bidder	+(+)	−(+)	−(+)
Rescue merger (good chaebol bidders acquiring bad targets in the same chaebol)	+(+)	−(+)	−(+)
Good performance/concentrated equity ownership/large size (high market value of equity)/business group	+	−	−

Panel B: Prediction and Summary Results of the Market Value Changes of Insider Holdings around the Merger Announcement Date

Bidder Characteristics	Prediction for the Market Value Changes of Insider Holdings for Bidder (Other Members of the Chaebol)		Empirical Results for the Market Value Changes of Insider Holdings for Bidder (Other Members of the Chaebol)
	Value-Added	Tunneling	
Insider holdings of chaebol bidders	+(+)	−(+)	−(+)

(b) Table II: value-added versus tunneling predictions.

7.2 Table III: Distribution of Merger Activity by Year and Bidder Industry and by Chaebol Affiliation and Industrial Relatedness; Table IV: Descriptive Statistics of Bidder Characteristics

Read: Table III shows that mergers are spread across years and concentrated in manufacturing, with variation in chaebol affiliation and relatedness. Table IV shows that Top 30 chaebol bidders differ from other bidders: they are larger, more levered, and have different governance and performance characteristics.

Economic message: These tables define the empirical sample and motivate the need for controls in the regression tests.

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Table III

Distribution of Merger Activity by Year and Bidder Industry and by Chaebol Affiliation and Industrial Relatedness

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We initially identify the sample from the *Securities Daily*, published by the KSE. We eliminate cases in which the acquiring firm owns all the shares of the acquired firm prior to the merger. The Top 30 chaebol bidders are firms that belong to 1 of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. Horizontal mergers are those in which the bidder and the target have the same three-digit industry classification codes provided by the KSE.

Panel A: By Year and Bidder Industry

Year	Construction	Manufacturing	Wholesale/ Retail	Transportation	Total
1981	1	6	0	0	7
1982	1	4	1	0	6
1983	1	5	2	0	8
1984	3	6	2	1	12
1985	4	5	0	0	9
1986	1	2	0	0	3
1987	1	5	0	0	6
1988	0	4	0	1	5
1989	0	2	0	1	3
1990	0	2	0	0	2
1991	0	5	0	0	5
1992	0	4	2	0	6
1993	0	5	0	0	5
1994	0	6	3	0	9
1995	1	7	1	0	9
1996	2	6	2	0	10
1997	0	2	0	0	2
Total	15	76	13	3	107

Panel B: By Chaebol Affiliation and Industrial Relatedness

Bidder and Target in the Related Industry | Bidder and Target

(a) Table III: merger activity by year, industry, chaebol affiliation, and industrial relatedness.

Table IV

Descriptive Statistics of Bidder Characteristics

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. The Top 30 chaebol bidders are firms that belong to 1 of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. We measure all variables at the fiscal year-end immediately before the merger announcement. We initially identify the sample from the *Securities Daily*, published by the KSE. We eliminate cases in which the acquiring firm owns all shares of the acquired firm prior to the merger. The numbers in the test-of-difference columns denote p-values.

	All Bidders N = 107		Top 30 Chaebol Bidders (A) N = 54		Top 31–50 Chaebol Bidders (B) N = 14		Other Bidders (C) N = 39		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Variables	Mean	Median	Mean	Median	Mean	Median	Mean	Median	t-test	Wilcoxon Z-test	t-test	Wilcoxon Z-test	t-test	Wilcoxon Z-test
Market value (in billions of won)	133.2	23.8	237.9	59.1	49.3	23.3	18.3	9.5	0.00***	0.08*	0.00***	0.00***	0.06*	0.01***
Total assets (in billions of won)	386.5	166.9	665.4	303.9	172.3	140.0	77.20	48.5	0.00***	0.00***	0.00***	0.00***	0.03**	0.00***
Debt/(debt + market value of equity)	0.75	0.79	0.74	0.75	0.74	0.81	0.77	0.84	0.92	0.80	0.56	0.46	0.62	0.30
Bank debt/(debt + market value of equity)	0.31	0.28	0.31	0.27	0.27	0.23	0.33	0.33	0.54	0.57	0.63	0.51	0.36	0.30
(Debt − bank debt)/(debt + market value of equity)	0.44	0.44	0.43	0.43	0.46	0.52	0.44	0.44	0.58	0.56	0.89	0.87	0.65	0.46
Equity ownership by the largest shareholder (a)	24.43	23.98	22.18	21.65	27.26	23.30	26.52	28.87	0.30	0.44	0.17	0.10*	0.90	0.84
Equity ownership by other corporations (b)	16.69	11.24	22.60	16.96	13.88	12.65	9.52	6.01	0.03**	0.13	0.00***	0.00***	0.23	0.10*
(a) + (b)	41.13	35.68	44.79	37.34	41.15	36.03	36.05	34.71	0.59	0.84	0.09*	0.68	0.38	0.51
Equity ownership by bank and insurance companies	9.93	6.70	14.81	12.32	4.46	2.75	5.14	3.23	0.00***	0.00***	0.00***	0.00***	0.69	0.85
Equity ownership by foreign investor	2.12	0.00	3.48	0.10	0.13	0.00	0.95	0.00	0.03**	0.00***	0.00***	0.00***	0.03**	0.95
Equity ownership by government	0.66	0.00	0.96	0.00	0.00	0.00	0.47	0.00	0.00***	0.02**	0.19	0.15	0.05**	0.12
Target book equity/bidder book equity	0.83	0.26	0.89	0.30	0.35	0.17	0.93	0.44	0.02**	0.09*	0.86	0.27	0.04**	0.01***
Prior bidder industry-adjusted stock return (−220 to −20)	0.01	0.01	0.01	−0.02	−0.12	−0.07	0.05	0.05	0.10*	0.28	0.55	0.18	0.05**	0.08*
Cash flow (operating income + depreciation)/total assets	0.10	0.07	0.12	0.07	0.08	0.08	0.07	0.08	0.69	0.61	0.29	0.66	0.49	0.64
Operating income/total assets	0.08	0.06	0.10	0.06	0.07	0.07	0.06	0.07	0.48	0.64	0.34	0.87	0.61	0.97
Financial slack (cash + marketable securities)/total assets	0.09	0.06	0.09	0.04	0.09	0.09	0.08	0.07	0.90	0.05**	0.90	0.12	0.72	0.51

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

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(b) Table IV: descriptive statistics of bidder characteristics.

7.3 Table V: Mean and Median Daily Abnormal Returns and Cumulative Abnormal Returns Around the Announcement Date

Read: Panel A reports daily abnormal returns; Panel B reports cumulative abnormal returns. The key pattern is that chaebol bidders, especially Top 30 chaebol bidders, experience negative announcement returns relative to other bidders.

Economic message: The bidder-value evidence rejects the simplest value-added story. The market does not treat chaebol acquisitions as good news for bidder minority shareholders.

Detailed interpretation: The event-study evidence is strongest because it uses short-run stock returns. Investors appear to infer that these mergers often transfer value away from the bidder rather than creating synergies.

Table V

Mean and Median Daily Abnormal Returns (AR) and Cumulative Abnormal Returns (CAR) for Bidders around the Announcement Date (AD)

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. The Top 30 chaebol bidders are firms that belong to one of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. AD denotes the initial announcement date. Numbers in parenthesis are *p*-values for the test that the mean/median is equal to zero.

Panel A: Daily Abnormal Returns (%) around the AD

	All Bidders		Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Day	Mean	Median	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
−10	0.207	−0.151	0.222	−0.282	−0.171	−0.866	0.323	−0.150	0.57	0.41	0.85	0.97	0.51	0.55
	(0.35)	(0.79)	(0.44)	(0.76)	(0.78)	(0.86)	(0.45)	(0.79)						
−9	0.178	−0.023	0.161	−0.032	0.691	0.207	0.017	−0.095	0.46	0.21	0.77	0.87	0.38	0.20
	(0.41)	(0.95)	(0.58)	(0.83)	(0.30)	(0.33)	(0.96)	(0.62)						
−8	0.294	0.024	0.306	0.129	−0.284	−0.547	0.485	0.021	0.36	0.43	0.80	0.76	0.37	0.64
	(0.31)	(0.89)	(0.31)	(0.57)	(0.62)	(0.67)	(0.46)	(0.98)						
−7	0.147	−0.090	−0.039	−0.184	1.172	0.130	0.036	−0.027	0.28	0.60	0.89	0.56	0.33	0.82
	(0.56)	(0.88)	(0.89)	(0.52)	(0.28)	(0.58)	(0.94)	(0.86)						
−6	0.423	0.050	0.108	−0.237	2.197*	1.283*	0.222	0.050	0.08*	0.07*	0.87	0.99	0.12	0.07*
	(0.18)	(0.46)	(0.75)	(0.95)	(0.06)	(0.06)	(0.71)	(0.96)						
−5	0.298	−0.066	0.185	−0.209	−0.804*	−0.651*	0.848	0.386	0.06*	0.21	0.30	0.39	0.02**	0.04**
	(0.25)	(0.87)	(0.52)	(0.97)	(0.07)	(0.06)	(0.14)	(0.23)						
−4	−0.090	−0.086	−0.147	−0.294	0.089	0.533	−0.075	−0.086	0.74	0.25	0.92	0.92	0.87	0.30
	(0.75)	(0.31)	(0.46)	(0.32)	(0.90)	(0.50)	(0.92)	(0.44)						
−3	0.200	−0.371	0.040	−0.147	−0.464	−0.455	0.659	−0.444	0.51	0.50	0.35	0.95	0.26	0.53
	(0.54)	(0.40)	(0.90)	(0.72)	(0.49)	(0.33)	(0.38)	(0.62)						
−2	0.506*	0.093	0.682**	0.149**	0.983	0.154	0.089	−0.200	0.74	0.86	0.32	0.06*	0.36	0.37
	(0.06)	(0.16)	(0.04)	(0.03)	(0.25)	(0.43)	(0.86)	(0.55)						

(a) Table V, Panel A: daily abnormal returns.

−1	0.692**	−0.027	0.193	−0.193	0.035	−0.332	1.619***	1.099***	0.86	0.92	0.04**	0.05**	0.09*	0.11
	(0.03)	(0.14)	(0.67)	(0.74)	(0.96)	(0.90)	(0.00)	(0.00)						
0	0.533	0.000	−0.051	0.027	0.438	−0.299	1.376	0.000	0.52	0.76	0.15	0.28	0.39	0.54
	(0.18)	(0.35)	(0.90)	(0.84)	(0.50)	(0.67)	(0.13)	(0.21)						
1	0.616*	−0.006	−0.527	−0.397	−0.243	−0.345	2.507***	2.080***	0.74	0.91	0.00***	0.00***	0.01***	0.03**
	(0.09)	(0.20)	(0.20)	(0.24)	(0.74)	(0.54)	(0.00)	(0.00)						
2	−0.058	−0.045	−0.733**	−0.427*	−0.760	−0.383	1.130*	0.725*	0.96	0.84	0.00***	0.01***	0.07*	0.06*
	(0.84)	(0.76)	(0.03)	(0.06)	(0.37)	(0.33)	(0.06)	(0.06)						
3	0.159	−0.098	−0.037	−0.181	0.839	−0.154	0.187	0.078	0.52	0.53	0.72	0.93	0.65	0.58
	(0.59)	(0.99)	(0.90)	(0.78)	(0.53)	(0.76)	(0.73)	(0.98)						
4	0.014	−0.202	−0.496*	−0.428*	1.417	0.915	0.216	0.023	0.15	0.09*	0.17	0.26	0.37	0.33
	(0.95)	(0.50)	(0.06)	(0.07)	(0.27)	(0.24)	(0.62)	(0.89)						
5	−0.203	−0.258	−0.163	−0.202	0.801	0.830	−0.618	−0.941*	0.27	0.22	0.37	0.17	0.13	0.10*
	(0.38)	(0.17)	(0.54)	(0.39)	(0.34)	(0.33)	(0.15)	(0.08)						
6	0.273	0.219	0.163	−0.033	1.470	0.378	−0.004	0.697	0.25	0.23	0.79	0.16	0.21	0.75
	(0.36)	(0.35)	(0.68)	(0.71)	(0.17)	(0.33)	(0.99)	(0.15)						
7	−0.204	−0.191	0.080	−0.062	−0.490	−0.576	−0.495	−0.204	0.53	0.26	0.37	0.32	0.99	1.00
	(0.44)	(0.38)	(0.74)	(0.92)	(0.58)	(0.54)	(0.40)	(0.38)						
8	−0.352	−0.226	−0.395	−0.462	−0.333	−0.195	−0.301	−0.016	0.93	0.86	0.86	0.92	0.96	0.98
	(0.14)	(0.16)	(0.24)	(0.27)	(0.59)	(0.81)	(0.48)	(0.56)						
9	0.015	−0.012	−0.247	−0.210	0.260	0.024	0.291	0.358	0.47	0.41	0.34	0.06*	0.97	0.49
	(0.95)	(0.88)	(0.47)	(0.12)	(0.67)	(1.00)	(0.52)	(0.19)						
10	−0.261	−0.132*	−0.199	−0.104	0.643	−0.030	−0.671*	−0.569**	0.38	0.63	0.31	0.17	0.19	0.25
	(0.25)	(0.10)	(0.46)	(0.56)	(0.48)	(0.95)	(0.08)	(0.05)						

Panel B: Cumulative Abnormal Returns (%) around the AD

	All Bidders		Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Event windows	Mean	Median	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
(AD − 10, AD − 2)	2.162*	0.415	1.520	0.501	3.408**	2.364*	2.605	−0.709	0.34	0.33	0.72	0.34	0.80	0.12
	(0.06)	(0.20)	(0.20)	(0.22)	(0.05)	(0.06)	(0.34)	(0.93)						
(AD − 1, AD)	1.225**	0.539**	0.141	−0.173	0.473	−0.591	2.995***	2.593***	0.79	0.81	0.03**	0.01*	0.09*	0.10*
	(0.02)	(0.04)	(0.84)	(0.90)	(0.64)	(0.90)	(0.01)	(0.00)						
(AD − 1, AD + 1)	1.841***	−0.086*	−0.386	−0.705	0.230	−0.371	5.502***	3.815***	0.67	0.60	0.00***	0.00***	0.00***	0.03**
	(0.00)	(0.09)	(0.64)	(0.26)	(0.85)	(0.95)	(0.00)	(0.00)						
(AD − 5, AD + 5)	2.666**	0.699	−1.054	−1.996	2.330	0.433	7.938***	4.130***	0.35	0.46	0.01***	0.00***	0.22	0.25
	(0.05)	(0.22)	(0.37)	(0.23)	(0.49)	(0.90)	(0.01)	(0.00)						
(AD − 10, AD + 10)	3.387**	1.682	−0.893	−0.331	7.485	1.914	7.842***	4.118***	0.16	0.40	0.04**	0.04**	0.96	0.61
	(0.05)	(0.11)	(0.60)	(0.78)	(0.19)	(0.42)	(0.04)	(0.02)						
(AD + 1, AD + 10)	0.000	−1.217	−2.554***	−1.995***	3.604	−0.968	2.241	1.989	0.17	0.48	0.03**	0.04**	0.77	0.74
	(0.99)	(0.39)	(0.01)	(0.01)	(0.40)	(0.63)	(0.26)	(0.29)						

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

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(b) Table V, Panel B: cumulative abnormal returns.

7.4 Figure 1: Cumulative Abnormal Returns from Day -20 to Day +20 Around the Merger Announcement; Table VI: Mean and Median CARs by Bidder Characteristics

Read: Figure 1 plots CARs for Top 30 chaebol bidders, Top 31–50 chaebol bidders, and other bidders. Top 30 chaebol bidders exhibit a clear negative path. Table VI shows that the negative effects are especially associated with bidder characteristics consistent with tunneling, such as strong bidders acquiring weak targets.

Economic message: The visual shows the timing of the value loss; the table shows which types of deals drive it.

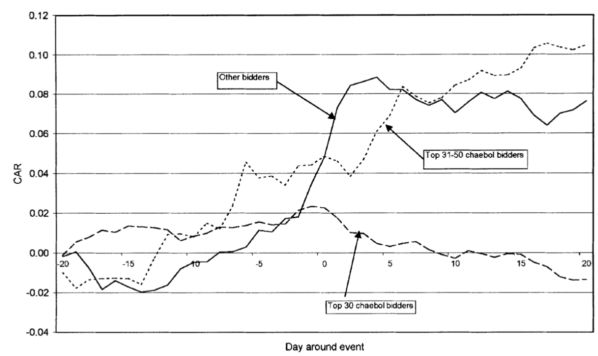

Figure 1. Cumulative abnormal returns from day −20 to day +20 around the merger announcement. The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. The Top 30 chaebol bidders are firms that belong to one of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms.

up its value before the acquisition announcement by having affiliated firms

Table VI

Mean and Median Cumulative Abnormal Returns [CAR(−5, 5)] for Bidders around the Merger Announcement Date for Subsamples Categorized by Bidder Characteristics

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. Numbers in parentheses are *p*-values for the test that the mean/median is equal to zero. Numbers in brackets are *p*-values for the test that the mean/median is equal between two groups of bidders.

Bidder Characteristics	No. of Firms	Mean CAR(−5, 5): % (*p*-value) [*t*-test for equality]	Median CAR(−5, 5): % (*p*-value) [Wilcoxon *Z*-test for equality]
Cash flow: (operating income + depreciation)/total assets			
Above sample median	53	0.506	−0.256
		(0.72)	(0.91)
Below sample median	54	4.786**	1.362*
		(0.04)	(0.07)
		[0.11]	[0.17]
Prior industry-adjusted stock return (−220 to −20)			
Above zero	55	1.436	−0.151
		(0.51)	(0.76)
Below zero	52	3.967***	1.740**
		(0.01)	(0.04)
		[0.35]	[0.09]*
Equity ownership of the largest shareholder and other corporation			
Above sample median	53	1.758	−0.256
		(0.72)	(0.91)
Below sample median	54	3.557**	1.362*
		(0.02)	(0.05)
		[0.51]	[0.13]
Market value of equity			
Above sample median	53	−0.798	−0.141
		(0.49)	(0.29)
Below sample median	54	6.066***	3.185**
		(0.01)	(0.02)

(a) Figure 1: cumulative abnormal returns around merger announcements.

(b) Table VI: CARs by bidder characteristics.

7.5 Table VII: Regression of CAR(-5,5) on Bidder Characteristics

Read: Table VII regresses bidder CARs on deal and bidder characteristics. The Top 30 chaebol dummy is negative and significant in most specifications. Cash flow is also negative, suggesting that strong bidders perform poorly when acquiring.

Economic message: Strong chaebol bidders appear to be used as vehicles for value transfer rather than value creation.

Table VII

Regression of Cumulative Abnormal Returns [CAR(−5, 5)] on Bidder Characteristics

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. The Top 30 chaebol bidders are firms that belong to one of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. The numbers in parentheses denote the standard error.

	All Bidders			
Independent Variables	(1)	(2)	(3)	(4)
Intercept	0.064	0.083	0.024	−0.017
	(0.141)	(0.138)	(0.165)	(0.162)
Dummy for the same industry	0.045*	0.055**	0.054**	0.067***
	(0.026)	(0.026)	(0.026)	(0.026)
Target is listed (dummy variable)	−0.024	−0.011	−0.003	0.003
	(0.029)	(0.029)	(0.031)	(0.030)
Target book equity/bidder book equity	0.034***	0.038***	0.039***	0.039***
	(0.011)	(0.011)	(0.011)	(0.010)
Log of market value of bidder equity	−0.002	−0.003	0.001	−0.003
	(0.009)	(0.008)	(0.010)	(0.010)
Book value of debt/(market value of equity + book value of debt)			0.027	0.058**
			(0.022)	(0.024)
Equity ownership of bidder by banks			−0.002	−0.001
			(0.002)	(0.002)
Equity ownership of bidder by foreign investors			0.002	0.000
			(0.003)	(0.003)
Dummy for Top 30 chaebol bidder	−0.091***	−0.079***	−0.072**	0.054
	(0.032)	(0.031)	(0.034)	(0.063)
Dummy for Top 31–50 chaebol bidder	−0.073*	−0.048	−0.052	0.077
	(0.043)	(0.042)	(0.043)	(0.080)
Prior bidder industry-adjusted stock return (−220 to −20)	−0.109**		−0.059	−0.040
	(0.045)		(0.049)	(0.048)
Cash flow (operating income + depreciation)/total assets		−0.159***	−0.259**	−0.247**
		(0.053)	(0.114)	(0.125)
Equity ownership of bidder by the largest shareholder and other corporations			−0.001	0.002*
			(0.001)	(0.001)
Equity ownership of bidder by the largest shareholder and other corporations × dummy for Top 30 chaebol bidder				−0.003**
				(0.001)
Equity ownership of bidder by the largest shareholder and other corporations × dummy for Top 31–50 chaebol bidder				−0.003
				(0.002)
Financial slack (cash + marketable securities)/total assets				−0.296
				(0.194)
Adjusted R^2	0.1905	0.2134	0.2243	0.2702
F-value	4.565***	5.108***	3.554***	3.617***
No. of observations	107	107	107	107

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

Table VII: cross-sectional regressions of bidder CARs on bidder and deal characteristics.

7.6 Table VIII: Mean and Median CARs Classified by Chaebol Affiliation and Return Performance; Table IX: Rescue Merger Tests

Read: Table VIII sorts returns by industry-adjusted prior performance and chaebol affiliation. Table IX focuses on rescue-merger settings, where good chaebol bidders acquire bad targets, especially within the same group.

Economic message: These tables sharpen the tunneling interpretation. If the value-added view were correct, strong bidders rescuing bad affiliates would not systematically harm bidder shareholders while helping other group members.

Table VIII

Cumulative Abnormal Returns [CAR(−5, 5)] Classified by Chaebol Affiliation and Prior Bidder Performance

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. The Top 30 chaebol bidders are firms that belong to 1 of the 30 largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. Numbers in parentheses denote *p*-values of the significance level. Numbers in brackets denote the number of observations.

Panel A: CAR (%) by Cash Flow (Operating Income + Depreciation) to Total Assets and Chaebol Affiliation

	Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other Bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Cash Flow to Total Assets	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
Above sample median	−4.256***	−4.387***	4.882	5.917	5.165**	1.090*	0.10*	0.03**	0.00***	0.00***	0.96	0.85
	(0.01)	(0.00)	(0.32)	(0.38)	(0.04)	(0.08)						
	[26]		[7]		[20]							
Below sample median	1.919	1.169	−0.222	−1.141	10.858*	4.837**	0.69	0.31	0.14	0.15	0.15	0.11
	(0.26)	(0.35)	(0.96)	(0.58)	(0.07)	(0.03)						
	[28]		[7]		[19]							
Test of difference												
t-test	0.00***		0.46		0.35							
Wilcoxon *Z*-test		0.01***		0.20		0.68						

(a) Table VIII, part 1: industry-adjusted bidder return classifications.

Panel B: CAR (%) by Industry-Adjusted Excess Returns and Chaebol Affiliation

	Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other Bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Industry-Adjusted Returns	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
Above zero	−3.825**	−3.388**	−1.932	−2.606	7.761*	0.847	0.60	0.67	0.02**	0.01***	0.09*	0.15
	(0.02)	(0.03)	(0.57)	(0.56)	(0.10)	(0.12)						
	[25]		[6]		[24]							
Below zero	1.335	−0.256	5.528	2.556	8.222**	4.837**	0.46	0.52	0.07*	0.04**	0.67	0.54
	(0.43)	(0.75)	(0.32)	(0.46)	(0.04)	(0.02)						
	[29]		[8]		[15]							
Test of difference												
t-test	0.02**		0.24		0.93							
Wilcoxon *Z*-test		0.06*		0.56		0.31						

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

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(b) Table VIII, part 2: continuation of industry-adjusted return classifications.

Table IX

Cumulative Abnormal Returns [CAR(−5, 5)] Classified by Chaebol Affiliation and Rescue/Nonrescue Mergers

The sample comprises 107 nonfinancial bidders listed on the KSE between 1981 and 1997. We obtain the initial public announcement date of the merger from the *Securities Daily*. We compute abnormal returns using the market model. We estimate the market model by using 200 trading days of return data ending 20 days before the merger announcement. We use the Korea Composite Stock Price Index (KOSPI) return as the benchmark. The Top 30 (50) chaebol bidders are firms that belong to one of the 30 (50) largest chaebols. The Top 31–50 chaebol bidders are firms that belong to chaebols ranked between 31 and 50. "Other bidders" are firms that belong to small chaebols or independent firms. Rescue mergers are those where the bidder acquires a target with a negative book value or negative income within the same chaebol. Numbers in parenthesis denote *p*-values of the significance level. Numbers in brackets denote the number of observations.

Panel A: CAR (%) by Rescue Merger and Chaebol Affiliation

	Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other Bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Rescue/Nonrescue	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
Rescue	−3.039*	−4.270*	−2.149	−1.212	6.720**	4.818*	0.84	0.69	0.01***	0.01***	0.11	0.31
	(0.09)	(0.07)	(0.62)	(0.87)	(0.04)	(0.07)						
	[17]		[4]		[19]							
Nonrescue	−0.142	−0.256	4.122	1.341	9.096*	2.713**	0.37	0.58	0.10*	0.05**	0.47	0.52
	(0.17)	(0.21)	(0.37)	(0.77)	(0.10)	(0.04)						
	[37]		[10]		[20]							
Test of difference												
t-test	0.21		0.31		0.69							
Wilcoxon *z*-test		0.22		0.52		0.99						

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(a) Table IX, part 1: rescue merger effects.

Panel B: CAR (%) of Chaebol Bidders by Cash Flow [(Operating Income + Depreciation) to Total Assets] and Rescue/Nonrescue Mergers

	Top 30 Chaebol Bidders (A)		Top 31–50 Chaebol Bidders (B)		Other bidders (C)		Test of Difference (A − B)		Test of Difference (A − C)		Test of Difference (B − C)	
Cash flow to total assets	Mean	Median	Mean	Median	Mean	Median	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test	*t*-test	Wilcoxon *Z*-test
Above sample median	−5.379**	−6.262**	−3.415	−4.072	−1.055	−1.183	0.75	0.76	0.17	0.15	0.71	0.71
	(0.04)	(0.05)	(0.58)	(0.75)	(0.63)	(0.50)						
	[8]		[3]		[22]							
Below sample median	−0.959	−3.104	1.648	1.648	2.367	0.737	—	—	0.32	0.46	—	—
	(0.71)	(0.65)	(—)	(—)	(0.27)	(0.50)						
	[9]		[1]		[25]							
Test of Difference												
t-test	0.20		—		0.26							
Wilcoxon *z*-test		0.22		—		0.39						

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

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(b) Table IX, part 2: continuation of rescue merger effects.

7.7 Table X: Regression of CARs for Chaebol Bidders and Group Portfolios; Table XI: Additional Tests of the Tunneling Effect

Read: Table X shows how bidder characteristics affect both bidder returns and the returns of other firms in the same group. Table XI reports additional tests using ownership, cash flow, market value, and insider-holding value changes.

Economic message: These tests are central to the paper's claimant-level logic: the bidder can lose while other group firms or insiders gain.

Detailed interpretation: The most important contribution of Table X and Table XI is to move beyond bidder returns. If only the bidder loses, one might interpret the acquisition as a bad investment. If other group firms or controlling shareholders gain, the pattern is much more consistent with tunneling.

Table X

Regression of Cumulative Abnormal Returns [CAR(−5, 5)] Realized by the Top 30 (50) Chaebol Bidders and the Portfolio of the Top 30 (50) Chaebol Firms that Belong to the Same Group as the Bidder on Bidder Characteristics

The sample comprises 51 (58) nonfinancial Top 30 (50) chaebol bidders, 51 (58) portfolios of Top 30 (50) chaebol firms excluding the bidders, and 51 (58) portfolios of Top 30 (50) chaebol firms including bidders during the years 1981 to 1997. We obtain the abnormal return for the value-weighted portfolio by estimating the market model parameters by using the return of the value-weighted portfolio. Good bidder/bad target dummy takes the value of one if the bidder's cash flow to total assets is above the sample median, the target's net income or book equity is below zero, and bidder and target are within the same business group. The numbers in parentheses denote the standard error.

	Top 30 Chaebol Bidders			Top 50 Chaebol Bidders		
Independent variables	Chaebol Bidder	Group Portfolio without Bidder	Group Portfolio with Bidder	Chaebol Bidder	Group Portfolio without Bidder	Group Portfolio with Bidder
Intercept	−0.097	0.109	0.090	−0.047	0.088	0.102
	(0.201)	(0.120)	(0.148)	(0.221)	(0.136)	(0.163)
Dummy for the related industry merger	0.041*	0.018	0.026	0.039	0.016	0.023
	(0.024)	(0.014)	(0.018)	(0.026)	(0.016)	(0.019)
Target is listed (dummy variable)	0.027	−0.056***	−0.027	0.022	−0.053	−0.028
	(0.027)	(0.016)	(0.020)	(0.030)	(0.018)	(0.022)
Target book equity/bidder book equity	0.016*	0.017***	0.017**	0.016	0.019***	0.018**
	(0.010)	(0.006)	(0.007)	(0.011)	(0.006)	(0.008)
Log of market value of bidder equity	0.006	−0.006	−0.004	0.003	−0.005	−0.004
	(0.009)	(0.005)	(0.007)	(0.010)	(0.006)	(0.007)
Prior bidder industry-adjusted stock return (−220 to −20)	−0.094*	−0.060*	−0.077**	−0.073	−0.056*	−0.072*
	(0.052)	(0.031)	(0.038)	(0.052)	(0.032)	(0.038)
Cash flow (operating income + depreciation)/total assets	−0.109***	0.006	−0.058**	−0.109***	0.001	−0.059*
	(0.039)	(0.023)	(0.028)	(0.043)	(0.026)	(0.032)
Bank debt/(market value of equity + book value of debt)	0.052	−0.146***	−0.045	0.060	−0.123**	−0.030
	(0.081)	(0.049)	(0.060)	(0.092)	(0.056)	(0.068)
(Total debt − bank debt)/(market value of equity + book value of debt)	−0.036	−0.035	−0.038	−0.037	−0.060	−0.061
	(0.083)	(0.050)	(0.061)	(0.092)	(0.056)	(0.068)
Equity ownership of bidder by banks	−0.001	0.001	0.000	−0.001	0.002	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Equity of ownership of bidder by foreign investors	0.002	0.000	0.000	0.002	0.000	0.000
	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
Equity ownership of bidder by the largest shareholder and other corporations	−0.001**	0.001**	−0.000	−0.001*	0.000	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Good bidder/bad target dummy	−0.065**	0.037*	−0.000	−0.063*	0.034	−0.000
	(0.033)	(0.020)	(0.024)	(0.037)	(0.022)	(0.027)
Adjusted R^2	0.3463	0.3268	0.2489	0.2261	0.1729	0.1458
F-value	3.208***	3.023***	2.381***	2.388**	1.993**	1.811*
No. of observations	51	51	51	58	58	58

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

G. Additional Tests of Tunneling

To check the robustness of the results, we conduct three additional tests.

Table X: CAR regressions for chaebol bidders and group portfolios.

Table XI

Additional Tests of the Tunneling Effect

The sample comprises 104 nonfinancial bidders that are listed on the KSE between 1981 and 1997. All bidders belong to a business group. The term "business group" denotes a group with at least two member firms. Insider holdings are shares held directly or indirectly by an owner-manager and his/her family. We obtain the market value changes of insider holdings by multiplying the insider holdings by the price change around the announcement date (−5 to 5). The Top 30 chaebol bidder is a firm that belongs to one of the 30 largest chaebols. The Top 31–50 chaebol bidder is a firm that belongs to a chaebol ranked between 31 and 50. Numbers in parentheses denote *p*-values of the significance level. Numbers in brackets denote the number of observations.

	Mean	Median
Panel A: CAR (%) by Sample Medians of Equity Ownership of the Largest Shareholder and Other Corporations and Market Value of Equity		
High equity ownership of the largest shareholder and other corporations/high market value of equity	−0.969	−1.185
	(0.64)	(0.54)
	[24]	
Low equity ownership of the largest shareholder and other corporations/low market value of equity	8.199***	5.810***
	(0.01)	(0.01)
	[24]	
p-value of difference		
t-test	0.01***	
Wilcoxon *z*-test		0.01***
Panel B: CAR (%) by Sample Medians of Cash Flow, Equity Ownership of the Largest Shareholder and Other Corporation, and Market Value of Equity		
High cash flow/high equity ownership of the largest shareholder and other corporations/high market value of equity	−5.770**	−4.714**
	(0.04)	(0.03)
	[11]	
Low cash flow/low equity ownership of the largest shareholder and other corporations / low market value of equity	9.084*	4.079*
	(0.07)	(0.08)
	[10]	
p-value of difference		
t-test	0.01***	
Wilcoxon *z*-test		0.01***

(a) Table XI, part 1: additional tunneling tests using ownership and firm characteristics.

	Market Value Changes of Insider Holdings					
Panel C: Market Value Changes of Insider Holdings in the Top 30 Chaebol Bidders (Million Won)						
	Top 30 Chaebol Bidders		Other Members of the Group		*p*-value of difference	
	Mean	Median	Mean	Median	*t*-test	Wilcoxon
Total sample (A)	−105	24	614	820	0.47	0.48
	(0.83)	(0.63)	(0.48)	(0.38)		
	[26]		[25]			
High cash flow (B)	−572	−75	760	1505	0.30	0.22
	(0.54)	(0.95)	(0.41)	(0.24)		
	[13]		[13]			
Low cash flow (C)	360	142	456	−381	0.95	0.76
	(0.37)	(0.50)	(0.77)	(0.97)		
	[13]		[12]			
p-value of difference (B − C)						
t-test	0.35		0.87			
Wilcoxon *z*-test		0.76		0.57		
Panel D: Market Value Changes of Insider Holdings in the Top 31–50 Chaebol Bidders (Million Won)						
Total sample (A)	−138	−215	−829	−495	0.59	0.53
	(0.73)	(0.62)	(0.50)	(0.31)		
	[5]		[5]			
High cash flow (B)	−130	−634	714	−289	0.56	0.38
	(0.89)	(1.00)	(0.59)	(1.00)		
	[3]		[3]			
Low cash flow (C)	−150	−150	−3142	−3142	0.03	0.25
	(0.26)	(0.50)	(0.04)	(0.50)		
	[2]		[2]			
p-value of difference (B − C)						
t-test	0.98		0.07			
Wilcoxon *z*-test		0.77		0.15		

*, **, and *** Significance at the 10, 5, and 1 percent levels, respectively.

(b) Table XI, part 2: insider-holding value changes and additional tests.

8 Short summary

- The paper tests value-added and tunneling views of Korean business groups.
- The sample contains Korean merger announcements from 1981 to 1997.
- Top 30 chaebol bidders experience negative announcement returns.
- The negative returns are stronger in settings consistent with tunneling.
- Strong chaebol bidders acquiring weak targets are especially important.
- Other firms in the same group and controlling shareholders can benefit even when bidder shareholders lose.
- The evidence supports the tunneling hypothesis more than the value-added hypothesis.

9 Conclusion

9.1 Core conclusion

Bae, Kang, and Kim show that mergers by Korean chaebol firms often reduce bidder shareholder value while increasing the value of other firms in the business group or the wealth of controlling shareholders. The evidence supports the view that mergers can be used as a tunneling mechanism.

9.2 Economic intuition

Chaebol controllers hold stakes across many firms and can exercise control that exceeds their cash-flow rights. If one affiliate is weak, a stronger affiliate can be used to rescue it. Minority shareholders of the bidder lose, but the controller may gain through ownership in the target, other group firms, or overall group survival.

9.3 Incremental contribution revisited

The paper's key contribution is to show that business groups in emerging markets can be both coordination devices and expropriation mechanisms. The event-study design allows the authors to separate value created for bidder shareholders from value shifted toward controlling shareholders and affiliated firms.

9.4 Broader implications

The findings imply that corporate governance, ownership concentration, and investor protection are essential for evaluating internal capital markets. Without strong minority-shareholder protection, internal transfers may be motivated by private benefits rather than efficiency.

9.5 Limitations

The evidence is based on Korean listed firms and merger announcements, and the study is not a randomized experiment. Still, the short-window return evidence and the group-level wealth-transfer tests provide strong support for the tunneling interpretation.

9.6 Takeaway

Business groups can look like internal capital markets, but they can also be tunnels. The difference depends on who controls the transactions and whose wealth is maximized.